

ARTICLE TYPE

Using Language Model to Predict Elections is an Undone Business

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Recent studies suggest that large language models (LLMs) can predict individual vote choices with high accuracy, ostensibly by inferring voters' underlying preferences. We reassess this claim using prompt-based ablations on American National Election Studies data from the 2012, 2016, and 2020 U.S. presidential elections. Replicating prior work, we observe high agreement between LLM predictions and reported votes when voter personas include party identification and ideological self-placement. However, performance drops sharply once these partisan cues are removed. Supplying policy platform information—whether generic, explicit, or emphasized—does not recover predictive accuracy. This pattern holds across model versions and election years. Taken together, our results indicate that the apparent success of LLM-based vote prediction largely reflects partisan label leakage rather than genuine inference about voters' policy preferences or political reasoning. We argue that predictive accuracy alone is insufficient evidence of substantive preference learning and emphasize the need to distinguish between prediction and inference when evaluating LLM-based methods in political analysis.

Keywords: Artificial Intelligence, Large Language Models, Elections, Predictions**1. Introduction**

Accurate election prediction matters for democratic accountability and for practical decisions by campaigns, media, and civic organizations, including how they allocate resources and respond to emerging electoral vulnerabilities. Yet generating reliable forecasts from voter data has become increasingly difficult. Survey-based polling faces persistent headwinds—declining response rates, rising fielding costs, and the challenge of reaching and weighting diverse populations—prompting greater reliance on multi-mode contact strategies and statistical corrections for nonresponse and other sources of bias.

In response, political forecasting research has increasingly embraced data-intensive approaches and statistical learning. Traditional forecasting models draw on historical fundamentals, demographics, and past electoral behavior; a prominent example is Abramowitz's time-for-change model (Abramowitz 2008). Bayesian approaches provide a complementary framework for incorporating uncertainty and updating predictions as new information arrives (Heidemanns, Gelman, and Morris 2020). More recently, machine learning methods have been used to detect predictive structure in rich voter datasets—for instance, tree-based models trained on demographic features to predict individual vote choice (Kim and Zilinsky 2021). Deep learning extends this approach by using multi-layer neural networks to model complex relationships in large-scale data, such as systems that infer election sentiment from Twitter during Pakistan's 2018 election (Ali et al. 2021). Related work in natural

language processing leverages textual signals from news and social media to estimate public sentiment and trends; for example, BERT-based models have been used to forecast the 2020 U.S. presidential election (Chandra and Saini 2021).

Large language models (LLMs) represent a further development in this trajectory. Unlike conventional forecasting pipelines, LLMs can be applied via prompting without task-specific retraining, can produce contextual rationales alongside predictions, and can potentially leverage broad linguistic and social knowledge acquired from large corpora. Recent work reports striking performance—up to 90

However, high accuracy alone does not reveal what information the model is using or whether it is learning substantive political preferences. If performance is driven primarily by partisan signals or their proxies embedded in the inputs, then apparent success may say more about label leakage than about genuine inference or political reasoning. Motivated by this concern, we investigate which components of voter personas and prompting strategies contribute to the performance of LLMs in election prediction. By identifying what does—and does not—drive accuracy, we aim to clarify the conditions under which LLM-based forecasting results are interpretable and reproducible, thereby providing a firmer foundation for future work.

2. Methods

We adopt the empirical framework of Argyle et al. (2023), which uses large language models (LLMs) to simulate individual-level vote choices from structured voter personas derived from survey responses. Our objective is not to develop new models, but to identify which information in the input personas drives predictive performance. Specifically, we test whether apparent forecasting success reflects substantive inference about voter preferences or instead relies on reconstructing partisan labels embedded in the prompt.

2.1 Data and Baseline Replication

Our analysis uses American National Election Studies (ANES) data from the 2012, 2016, and 2020 U.S. presidential elections. Following Argyle et al. (2023), we constructed simulated voter personas from a subset of individual-level variables, including demographic characteristics, political interest, and political attitudes. We then provided these personas to the language model as textual prompts and asked it to generate a binary vote choice for the major-party candidates in each election year.

For the baseline, we used the original prompt specification, which includes respondents’ party identification and ideology. This replication provides a benchmark against which we compare alternative prompt designs.

2.2 Prompt-Based Ablation as an Identification Strategy

Our identification strategy relies on systematic prompt-based ablation, in which we selectively remove or replace information available to the model. These ablations function as information-exclusion tests: if predictive performance remains high after a piece of information is removed, that information is not necessary for the model’s predictions. Conversely, a substantial drop in performance indicates that the model relies on that information.

Our primary ablation removes explicit partisan cues—party identification and ideology—from the voter personas. In modern U.S. elections, partisanship and ideology are strong predictors of vote choice, but they operate as summary labels rather than substantive explanations. Removing them allows us to assess whether the model can infer vote choice from remaining traits or instead depends on direct partisan shortcuts.

In addition to the variables used by Argyle et al. (Argyle et al. 2023), we incorporate candidates’ policy platforms into the prompts, motivated by evidence that policy positions shape electoral outcomes (Kulachai, Lerdtomornsakul, and Homyamye 2023). To evaluate whether policy-based

reasoning can substitute for partisan labels, we design three prompt conditions that exclude party identification and ideology: (1) instructing the model to consider each candidate’s platform; (2) providing explicit policy summaries drawn from public sources (e.g., The American Presidency Project); and (3) prompting the model to identify key policy differences before predicting the vote. Together, these conditions test whether policy information alone can recover predictive accuracy.

Table 1. Examples of prompt conditions used in GPT simulations. Prompt examples corresponding to each experimental condition, illustrating how information about party identification, ideology, and policy platforms was manipulated to assess their contribution to model predictions.

Condition	Example
0a. With Party Identification and Political Ideology (Argyle et al., 2023)	Racially, I am black. Ideologically, I am extremely liberal in my views. Politically, I am a strong Democrat. I attend church. I am 86.0 years old. I am a woman. I am from Alabama. Answer this in two words: In the 2012 presidential election, I voted for...
0b. With Party Identification and Political Ideology (Replication)	(ibid.)
1a. Without Party Identification	Racially, I am black. Ideologically, I am extremely liberal in my views. I attend church. I am 86.0 years old. I am a woman. I am from Alabama. Answer this in two words: In the 2012 presidential election, I voted for...
1b. Without Political Ideology	Racially, I am black. Politically, I am a strong Democrat. I attend church. I am 86.0 years old. I am a woman. I am from Alabama. Answer this in two words: In the 2012 presidential election, I voted for...
1c. Without Party Identification and Political Ideology	Racially, I am black. I attend church. I am 86.0 years old. I am a woman. I am from Alabama. Answer this in two words: In the 2012 presidential election, I voted for...
2a. Implicit Political Platform	Racially, I am black. I attend church. I am 86.0 years old. I am a woman. I am from Alabama. Answer this in two words: <i>After considering the positions of all the candidates in the 2012 presidential election, I voted for...</i>
2b. Explicit Political Platform	<i>In the 2012 presidential election, President Barack Obama’s platform focused on economic recovery, tax cuts for middle-class families, investment in education and clean energy, and healthcare reform through the Affordable Care Act. He emphasized comprehensive immigration reform, including a pathway to citizenship, and sought to improve education through increased funding and accountability. Mitt Romney’s campaign, on the other hand, proposed tax and entitlement reforms to stimulate economic growth, advocated for repealing the Affordable Care Act in favor of state-driven solutions, and promoted school choice and strict immigration enforcement to bolster economic and educational outcomes.</i> Racially, I am black. I attend church. I am 86.0 years old. I am a woman. I am from Alabama. Answer this in two words: <i>After considering the positions of all the candidates in the 2012 presidential election, I voted for...</i>
2c. Recalled Political Platform	<i>In the 2012 presidential election, the most memorable policy positions were President Barack Obama’s focus on healthcare reform through the Affordable Care Act and Mitt Romney’s emphasis on reducing the federal deficit and repealing the Affordable Care Act.</i> Racially, I am black. I attend church. I am 86.0 years old. I am a woman. I am from Alabama. Answer this in two words: <i>After considering the positions of all the candidates in the 2012 presidential election, I voted for...</i>

2.3 Model Implementation and Reproducibility

We evaluate two model versions: GPT-3.5-turbo-0125 and GPT-4o. All queries are run with the temperature parameter set to zero to minimize stochastic variation in generated responses. This setting is chosen to support identification and reproducibility rather than to optimize headline accuracy. By forcing deterministic outputs, observed differences across prompt conditions can be attributed to the information provided in the persona rather than to sampling noise in generation.

For each simulated voter, we submit a single prompt and record the model’s predicted vote

choice. We exclude invalid or nonresponsive outputs using the same criteria as (Argyle et al. 2023). Predicted vote choices are then merged back into the corresponding American National Election Studies (ANES) records for analysis.

2.4 Evaluation Metrics

We assess predictive performance using two complementary measures. First, we calculate the tetrachoric correlation between predicted and the observed vote choices. This metric is appropriate for binary outcomes and can be interpreted as the correlation between latent continuous propensities underlying the observed dichotomous variables. Second, we report agreement, defined as the proportion of cases in which the model’s prediction matches the respondent’s reported vote.

Our focus is comparative rather than maximization-oriented. We use changes in these metrics across ablation conditions to diagnose how sensitive LLM vote predictions are to the inclusion or removal of specific information in the voter persona.

3. Results

Across election years and model versions, we observe a consistent qualitative pattern: high predictive performance emerges only when voter personas include explicit partisan cues, and policy-focused prompting fails to recover accuracy once these cues are removed.

3.1 Replication of Prior Findings

We begin by replicating the baseline prompt specification in (Argyle et al. 2023), which includes both party identification and ideological self-placement in the voter persona. Under this condition, both GPT-3.5 and GPT-4o closely match observed ANES vote choices in 2012, 2016, and 2020. Tetrachoric correlations exceed 0.95 in each election year, with an approximate agreement of 0.90.

These results closely track prior reports, indicating that our implementation reproduces the headline finding of high LLM-based vote-prediction accuracy when partisan identifiers are available.

3.2 Removing Party and Ideology Collapses Predictive Accuracy

The central empirical result emerges once party identification and ideological self-placement are removed from the persona. Under this ablation, predictive performance drops sharply for both models in all three election years.

Relative to the baseline specification, tetrachoric correlations decline substantially, and agreement falls to levels only modestly above chance. In other words, without direct partisan signals, the models do not reliably recover vote choice from the remaining demographic and attitudinal attributes alone.

This degradation is consistent across model versions. GPT-4o occasionally yields slightly higher agreement than GPT-3.5, but these differences are minor compared to the overall loss induced by removing party and ideology. The similarity across models suggests that the reliance on partisan identifiers reflects a general feature of the prompt-based simulation setup rather than a limitation of any single model.

3.3 Policy-Based Prompting Does Not Recover Accuracy

We next test whether policy-related information can compensate for the absence of partisan labels. We examine three variants: (i) a generic instruction to consider the candidates’ platforms, (ii) explicit summaries of each candidate’s policy positions, and (iii) prompts emphasizing salient policy differences (i.e., a "recalled" platform framing) before predicting the vote choice.

Across all three conditions, predictive performance remains low. Neither tetrachoric correlations nor agreement returns to the baseline levels achieved when party and ideology are included. In many cases, results are similar to—or only marginally better than—those obtained when party and

ideology are omitted. Even when modest gains appear in particular years or for a specific model, they are not consistent across elections and do not meaningfully close the gap created by withholding partisan identifiers.

Overall, providing policy information does not substitute for party identification and ideological self-placement as the basis for accurate vote-choice prediction in these simulations.

3.4 Summary of Empirical Patterns

Taken together, the evidence points to a straightforward conclusion: prompt-based LLM election prediction performs well when explicit partisan identifiers are present, but breaks down when those cues are withheld. Moreover, adding policy-focused instructions or policy summaries does not restore the original level of accuracy. This pattern is consistent with an interpretation in which the apparent success of LLM-based vote prediction is primarily driven by partisan label information in the prompt, rather than by substantive inferences about voters' policy preferences or political reasoning.

These results are summarized visually in **Figure 1**; prompt examples are provided in **Table 1**.

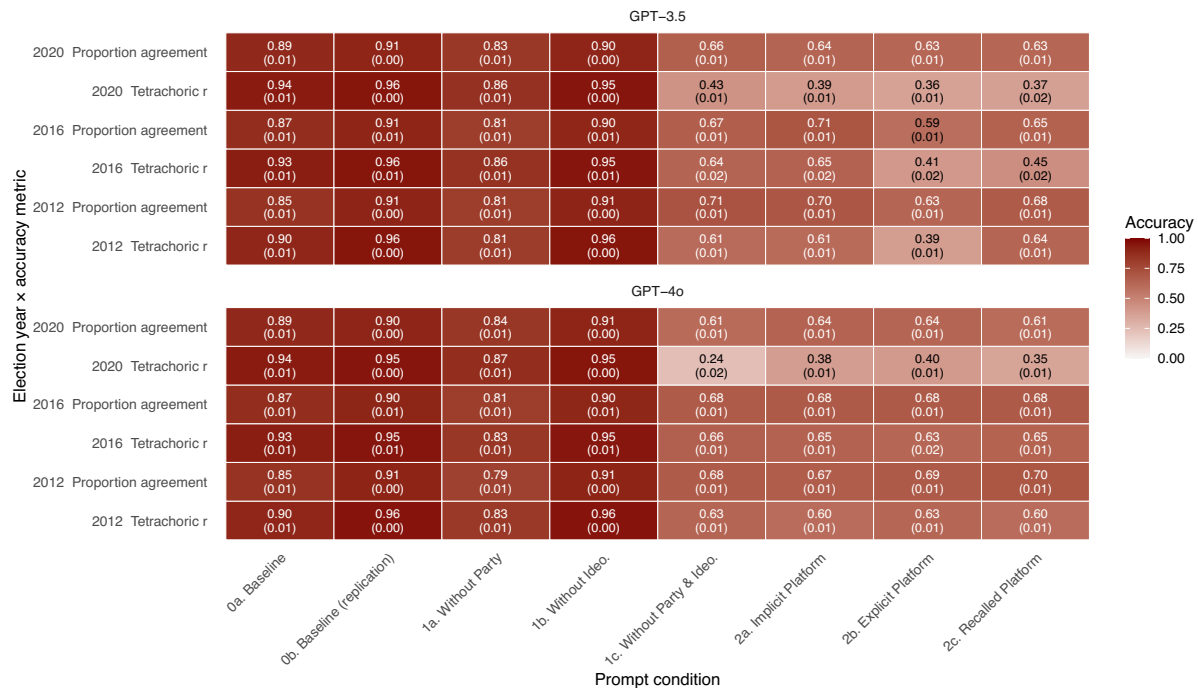


Figure 1. Accuracy of GPT predictions across prompt conditions and election years. Heatmap displaying predictive accuracy (tetrachoric correlation and proportion agreement) for each election year (2012, 2016, 2020) across prompt variants. Accuracy is highest when party identification and ideology are included (0a/0b), and drops significantly when they are removed (1a–1c), with policy-based prompts (2a–2c) failing to recover performance. Values are shown with standard errors in parentheses.

4. Discussion & Conclusions

In this study, LLM-based vote-choice prediction appears highly sensitive to whether voter personas include explicit partisan cues. When party identification and ideological self-placement are removed, predictive accuracy drops sharply across years and models. Policy-focused prompting—whether by instructing the model to consider candidates' platforms, supplying platform summaries, or prompting recall of salient election-year policy differences—produces little improvement. This is especially evident for GPT-4o, where none of the three policy variants meaningfully restores performance.

These findings temper strong claims about LLMs' ability to infer individual political preferences from limited survey-style descriptions. Prior work reports high accuracy in predicting vote choice. However, our results indicate that this accuracy depends heavily on the inclusion of party and ideology, rather than emerging from broader preference inference. This is consistent with the broader observation that strong performance on general cognitive benchmarks does not necessarily translate into reliable prediction in complex, socially embedded behaviors such as voting Atari *et al.* 2023.

At the same time, we observe pockets of partial recovery in specific contexts. For example, in the 2012 election, adding policy information increases GPT-3.5's tetrachoric correlation from very low levels to approximately 0.6, similar to GPT-4o under the same condition. Although this remains well below the baseline condition that includes party identification, it suggests that policy information can sometimes provide a usable signal, depending on the electoral context and the population subgroup being simulated.

These patterns also help clarify why policy prompting alone may be insufficient in our setup. First, once party and ideology are removed, the remaining persona attributes may be too sparse to support accurate individual-level prediction. Platforms are only one potential input; voter decision-making is also shaped by socioeconomic circumstances, social networks, media environments, regional and community contexts, social identities (e.g., race and religion), and social-psychological factors such as values and personality traits. Incorporating richer measures of these factors—or modeling how they interact with policy positions—may be necessary to test whether LLMs can move beyond partisan cue reconstruction toward more substantive preference inference.

Despite these limitations, LLM-based approaches remain worth investigating, provided that evaluations are designed to distinguish between prediction and inference. Future work could examine when and how supplementary information (including policy descriptions, media cues, or local context) improves performance, and whether gains generalize across elections and groups rather than appearing only in isolated cases. At the same time, any move toward operational forecasting or intervention-adjacent applications raises ethical concerns, including risks of strategic manipulation and unequal influence. Progress in this area should therefore pair methodological development with careful attention to interpretability, robustness, and governance.

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Conflicts of Interest None

Data Availability Statement All the data and codes in our analysis are available on the project's Open Science Framework page: <https://doi.org/10.17605/OSF.IO/U32ZK>.

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